

Introduction to Statistical Methods in Data Science

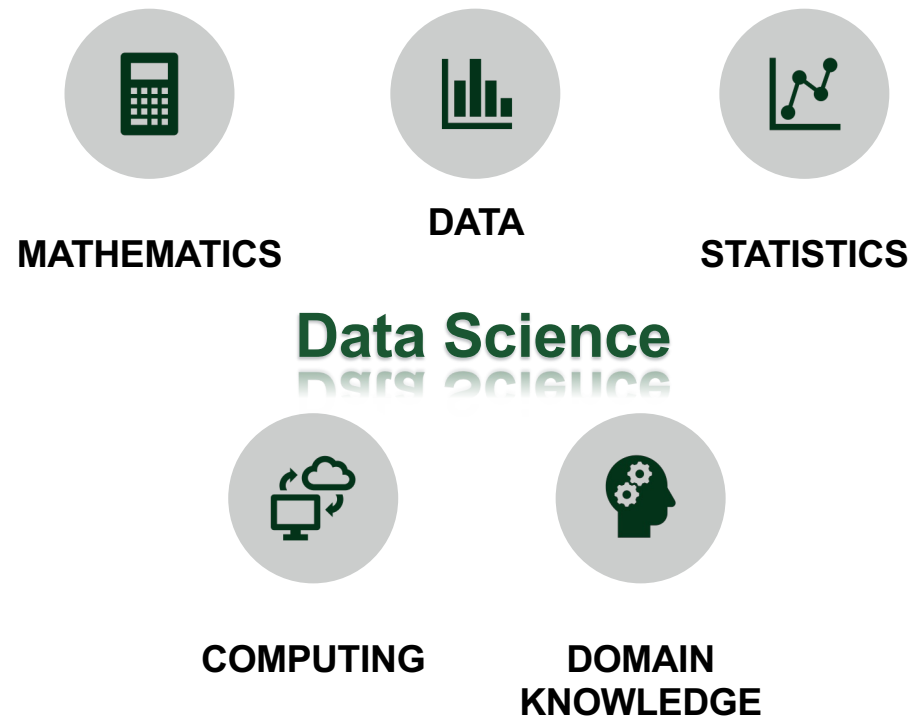
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University of Alabama at Birmingham
@ Data Science UAB Summer Camp

What is Data Science?

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Examples:

- Netflix recommendations
- Weather forecasting
- Medical diagnosis
- Sports analytics



Why Data Matters

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Examples:



Google Search



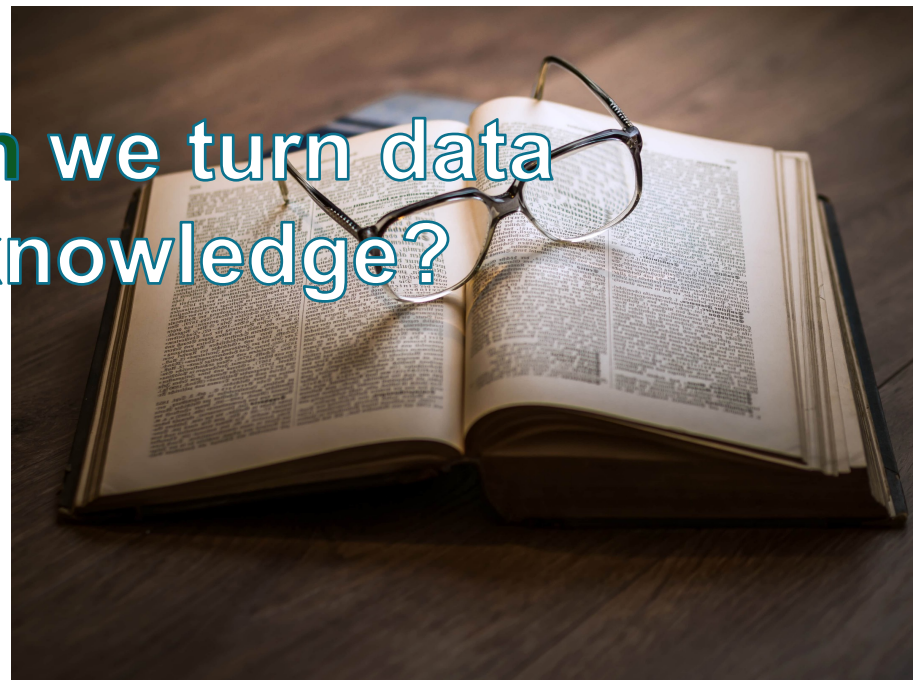
Amazon Purchase



Smartwatch Measurement

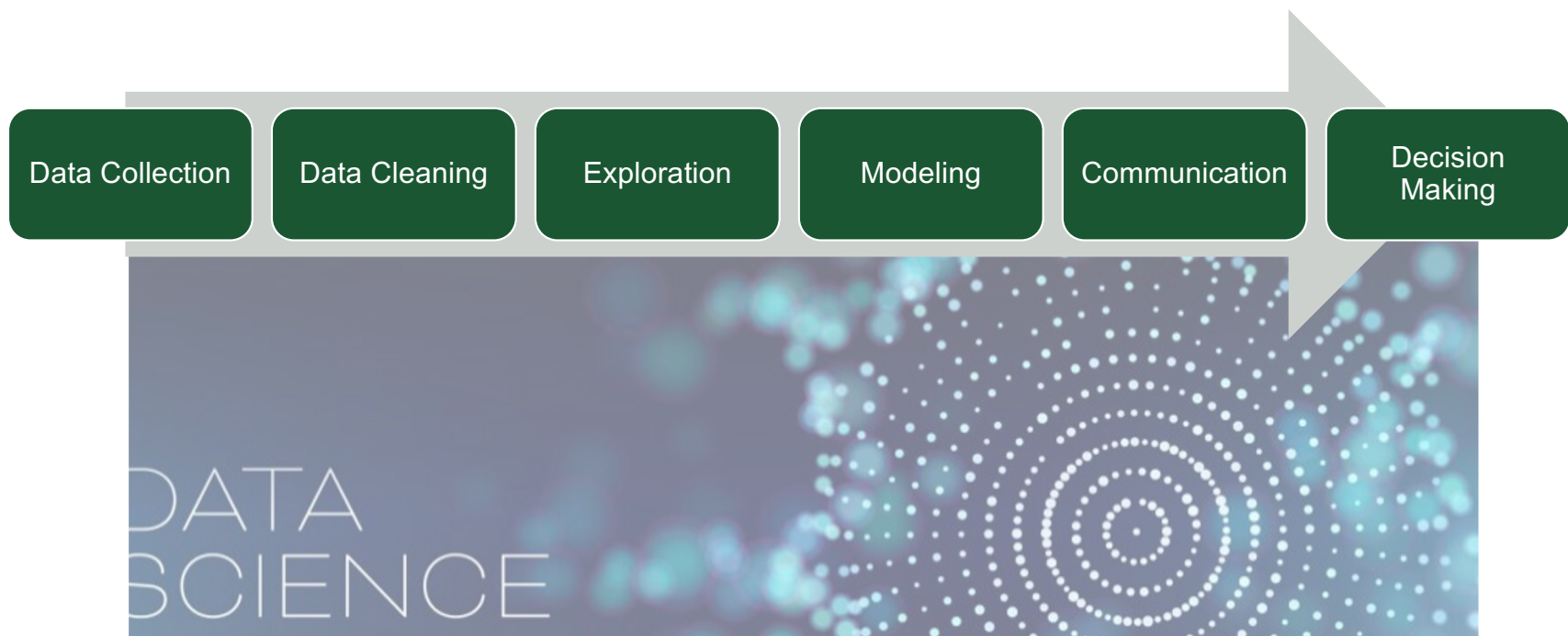
Millions every day

**How can we turn data
into knowledge?**



The Data Science Workflow

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What Does Statistics Do?

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What happened?

Why did it happen?

What might happen next?

How certain are we?

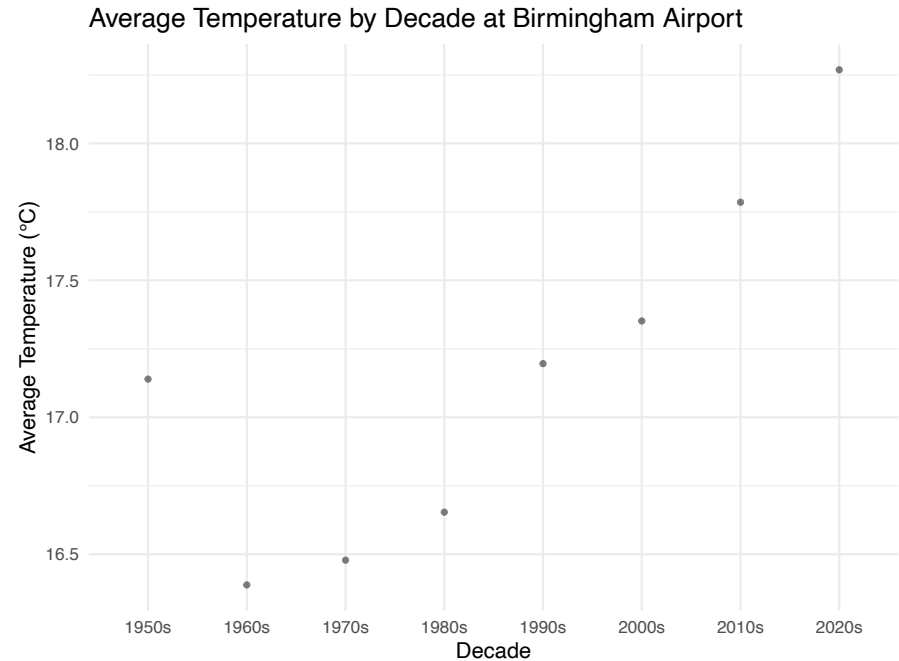
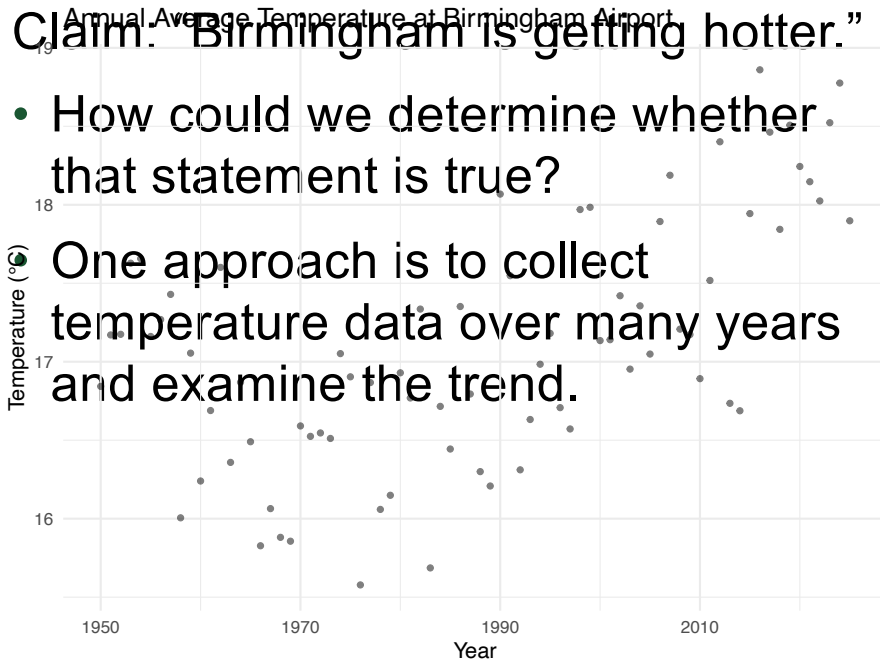
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Example: Is Birmingham Getting Hotter?

Claim: "Birmingham is getting hotter."

- How could we determine whether that statement is true?

One approach is to collect temperature data over many years and examine the trend.



Population vs Sample

Population



All high school students in Alabama

Sample



Students in this room

Can this sample tell us something about all Alabama students?

Statistics uses samples to learn about populations.

Data Types

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Which type of data
does ChatGPT use
most?

Categorical Data

-  School
-  Gender
-  Car Brand

Numerical

-  Height
-  Test Score
-  Income

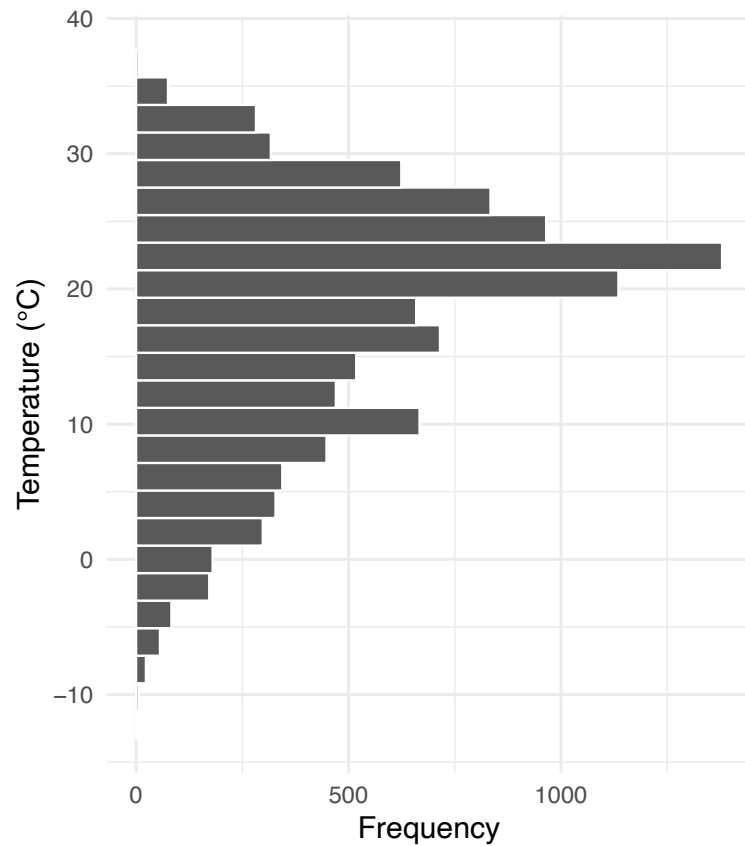
Images:

-  Photo
-  X-Ray
-  Self-Driving Car Camera

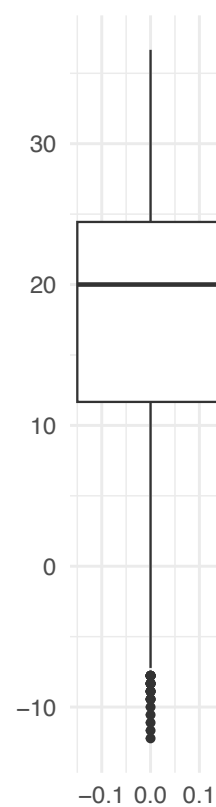
Text:

-  Social Media Posts
-  Emails
-  News Articles

Birmingham Temperatures 2025



Boxplot



Observations: 10586

Mean: 17.9 °C

Median: 20.0 °C

Minimum: -12.2 °C

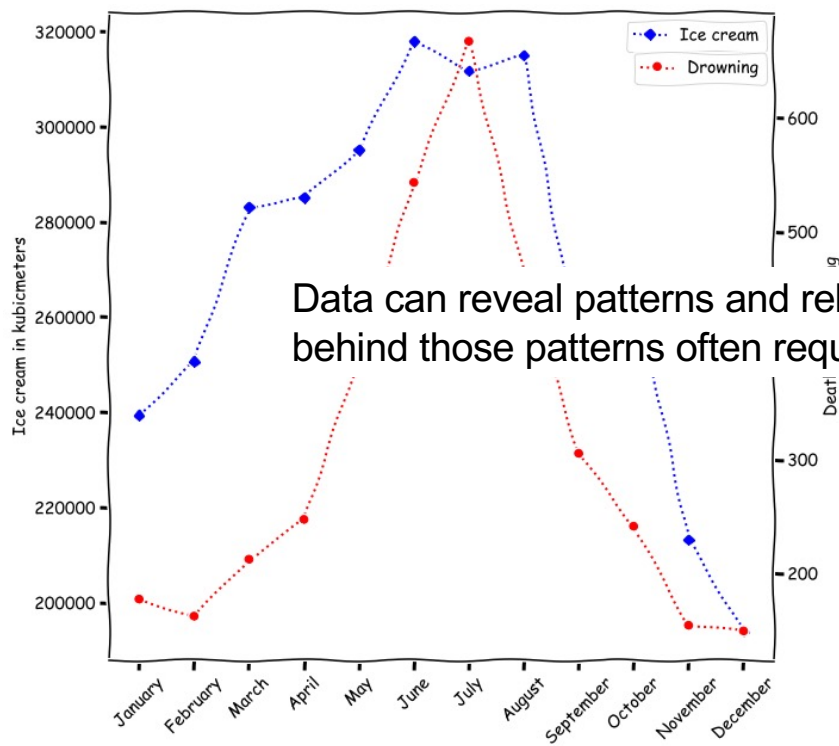
Maximum: 36.7 °C

Summarizing Data

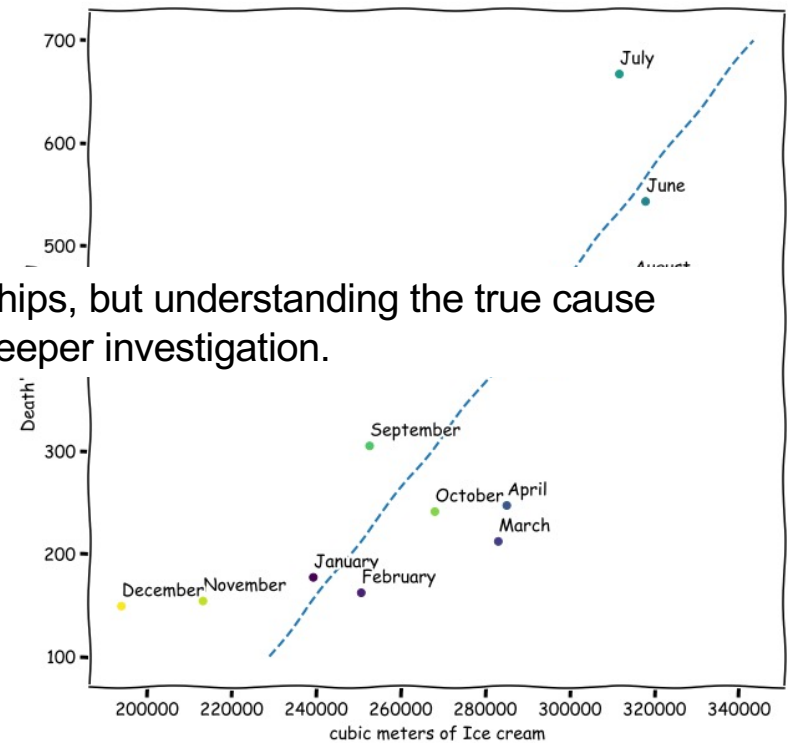
Statistics helps us transform large collections of observations into understandable summaries and insights.

Ice Cream Sales vs Drowning

Correlation does not mean causation!



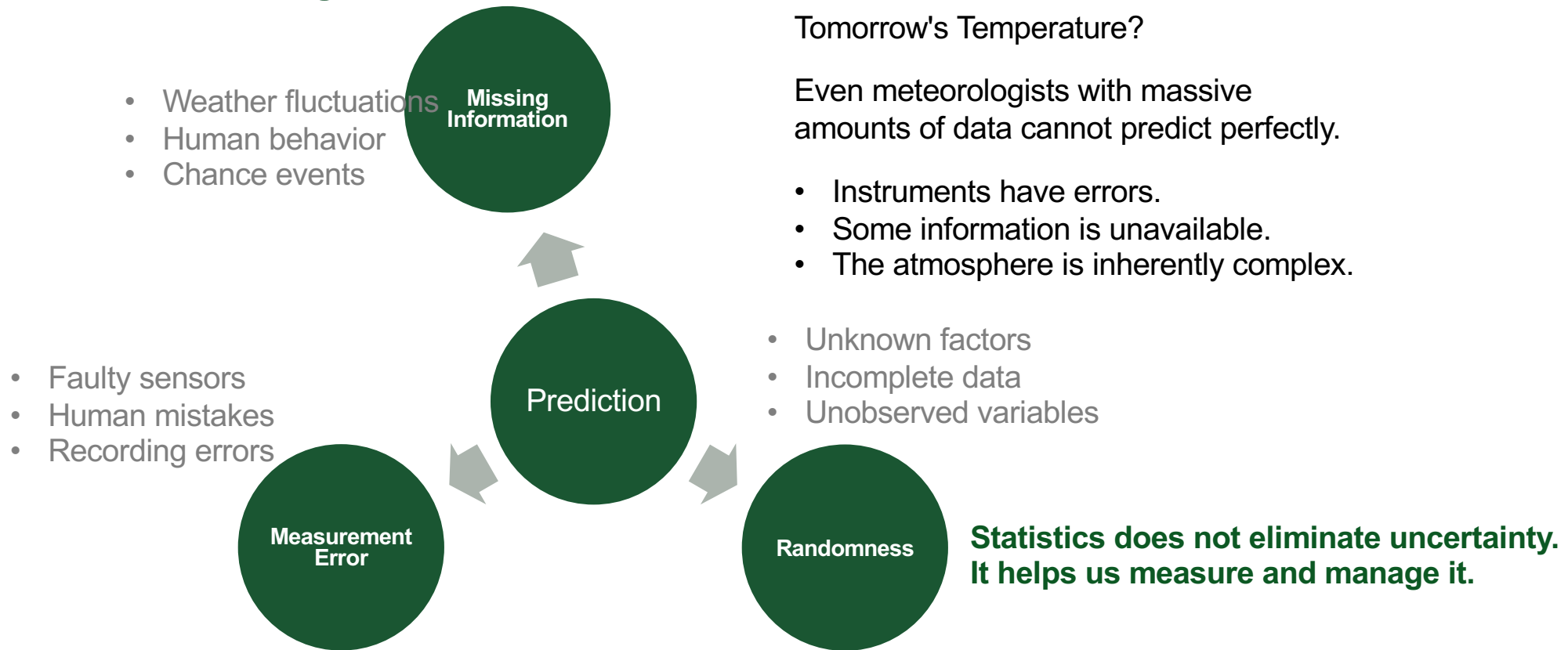
Data can reveal patterns and relationships, but understanding the true cause behind those patterns often requires deeper investigation.



Source: Andreas R. Madsen (2020), "A Story of Ice Cream, Drowning, and Causal Modelling"

Uncertainty

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Data Science in Action

Data science helps solve real-world problems.

Healthcare

Disease prediction
Medical imaging
Personalized treatment



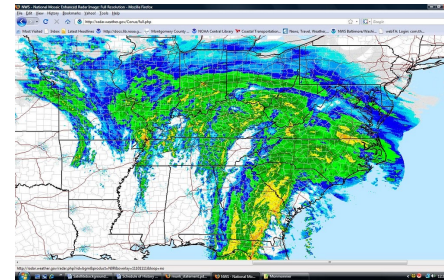
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Finance

Credit card fraud detection
Stock market analysis
Loan approval

Climate Science

Weather forecasting
Climate change studies
Natural disaster prediction



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Social Media

TikTok recommendations
YouTube suggestions
Content moderation

Key Takeaways

Statistics helps us:

- Understand data
- Quantify uncertainty
- Support decision making

Data alone is not knowledge. Statistics helps transform data into knowledge.

Questions?

